

Efficient Crawling for Scalable Web Data Acquisition

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Ever increasing volume of content of many forms on the Web:

- Corpora of text and video → Training of ML or DL models
- Structured HTML (especially containing tables) → Knowledge Bases
- Open-access statistic dataset and studies → Analyzing how societies function and doing fact-checking.
- ...

First step for building applications based on Web-published data is to find those data files.

Context (2/3)

Data files come in a wide variety of **formats**:

- **Text** and **video** → .doc, .txt, .mp4, .flv...
- **Structured HTML** → .html
- **Statistic dataset and studies** → .pdf, .xls, .csv...

They are published by different **providers**:

- *National agencies*: INSEE (France), Labor/Census Bureaus (US), ABS (AU)
- *International bodies*: Eurostat, IMF, WB, UN Agencies → some provide **APIs or portals**
- *Smaller entities*: Trade unions, NGOs → publish **highly-specialized data**

Websites can be huge with only a small fraction of published data files

Exhaustively crawling those websites takes a lot of **space** and most importantly **time**:

- Taken by the crawl of a lot of “uninteresting” resources
- Crawling ethics requires to wait (typically ~ 1 sec.) between two HTTP requests: for a website of **1M pages** $\rightarrow$ roughly **11 days** of waiting...

Generic crawler for any given website (selected beforehand) $\rightarrow$ we **cannot make any assumption about the website's structure/content**.

Problem Statement

Problem: Given the starting URL of a website, we want to **retrieve** as many **targets** as possible, while **minimizing** the consumption of **resources**. We define:

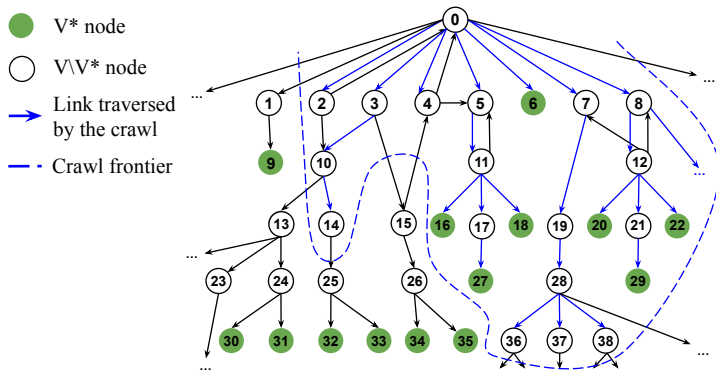
- A **target**: file uploaded on the website, satisfying some constraints specified by the user (e.g., be a data file: CSV, Excel, PDF...).
- **Resources**: number of **HTTP requests** sent to the server, and **data volume** exchanged with the server.

We want to retrieve the maximum of targets while minimizing the effort to do so.

Formalization: Graph Crawling Problem (1/2)

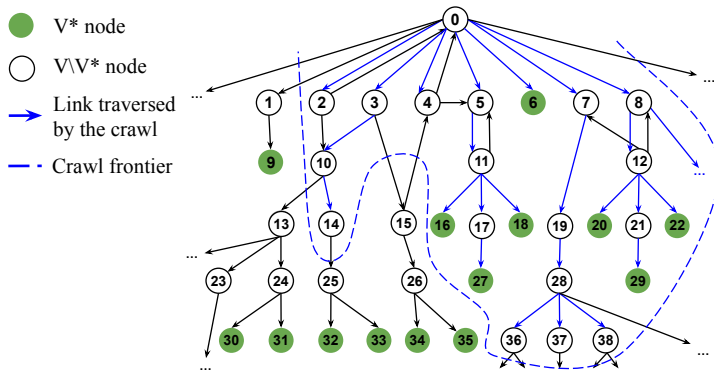
Website graph: $G = (V, E, r, \omega, \lambda)$, with:

- V a finite set of *vertices* (webpages)
- $E \subseteq V^2$ a set of *edges* (hyperlinks)
- $r \in V$ the *root* of the graph (input webpage; 0 here)
- $\omega : V \rightarrow \mathbb{R}^+$ a *cost* function (cost of retrieving that page)
- $(\lambda : E \rightarrow \mathcal{L}$ a *labeling* function).



Formalization: Graph Crawling Problem (2/2)

Given a website graph $G = (V, E, r, \omega, \lambda)$ and a subset V^* of **targets** of V , the **graph crawling problem** is to find a crawl $T = (V', E')$ of G with $V^* \subseteq V'$, of minimal total cost.



NP-hard problem (reduction from **set cover**): thus needing a **heuristic** yet **effective** solution to solve our crawling problem

Existing Web Crawlers (1/3)

What are **existing Web crawlers** from the literature?

Most simple approaches:

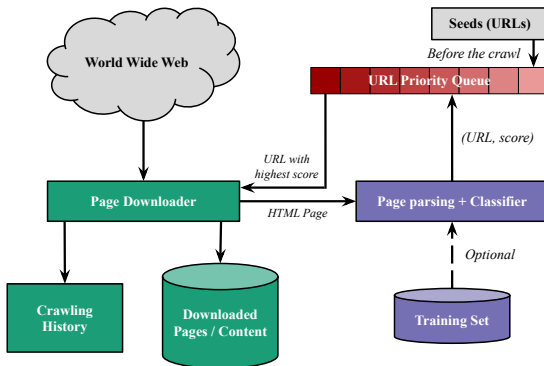
- **Random Crawler**: selects the next hyperlink to visit uniformly at random from the frontier. **Most naive technique**.
- **Breadth-First Search (BFS)**: frontier as a **First-In-First-Out** queue data structure. Efficient when targets are not too deep.
- **Depth-First Search (DFS)**: frontier in a **Last-In-First-Out** stack.

None of them is effective without prior knowledge of the **website's structure**, **which we don't have**

Existing Web Crawlers (2/3)

Focused Crawlers:

- Prioritize some pages during a crawl, following a **focus** (an objective).
- Usually, a **classifier** assesses the **likelihood** that an hyperlink leads to a page part of the focus.
- **Frontier**: a *priority-queue*.



Existing Web Crawlers (3/3)

Two kind of focused crawlers:

- **Topical**: **focus** is a predefined **topic** $\rightarrow$ keywords or sample documents.
- **Non-Topical**: rarer; **focus** is something else:
 - ANTHELION [MMB14]: semantic annotations in Web pages
 - ACEBOT [FS15]: as much diverse textual content as possible

In our setting: topical crawlers are **ill-suited**

- We want targets about **any possible subject** covered by the website
- Defining a topic $\implies$ choosing **one (or several) languages**

Best approach: **non-topical focused crawlers**

Research Hypotheses

Our approach → two main research hypotheses:

- Hyperlinks similarly structured in the HTML page were they were found lead to similar content.
- It is possible to learn which hyperlinks are most likely to lead to targets, given their link structure.

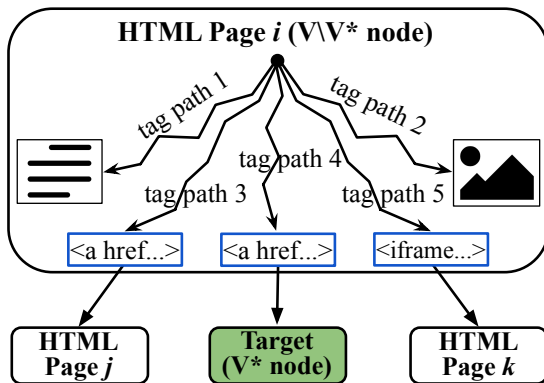
Idea: We can separate, for a given website, parts of it that are rich in targets, from ones that are not:

- without prior knowledge of the website's structure/content
- just assuming existence of some structure

Building Groups of Hyperlinks

How to do it: We represent each encountered hyperlink with its associated **tag path** (its **link structure**).

They are computed from the HTML page's **Document Object Model**:



Building Groups of Hyperlinks

We form **groups of hyperlinks** with a merging strategy based on **similarity** between the tag paths...

...to **separate the fruitful groups** from the less interesting ones, as we assume that **similar hyperlinks will lead to similar kind of content**.

```
https://opm.justice.gouv.fr/content/notice-dépôt-dune-candidature-au-concours-de-greffier-de-tribunal-de-commerce-annexe-1-0  
/html/body.html front not-logged-in one-sidebar sidebar-first page-node/div.jumbotron opm-content-container/div.container/div.row/div.col-sm-3 well-resp/div.block block-block last odd#block-block-7/p/a
```

```
https://www.justice.gouv.fr/lettre-direction-affaires-civiles-du-sceau
```

```
/html/body.html front not-logged-in one-sidebar sidebar-first page-node/div.jumbotron opm-content-container/div.container/div.row/div.col-sm-3 well-resp/div.block block-block first odd#block-block-6/p/a
```

```
https://opm.justice.gouv.fr/content/notice-de-renseignements-sur-l'accès-à-la-profession-de-greffier-de-tribunal-de-commerce-et
```

```
/html/body.html front not-logged-in one-sidebar sidebar-first page-node/div.jumbotron opm-content-container/div.container/div.row/div.col-sm-3 well-resp/div.block block-block last odd#block-block-7/p/a
```

```
https://www.justice.gouv.fr/grands-dossiers/justice-amiable  
/html/body.path-frontpage page-node-type-homepage/[...] /div.fr-col-md-6fr-col-12/article.fr-enlarge-link fr-card fr-card-detail-sm fr-card-horizontal/div.fr-card_body/div.fr-card_content/h3.fr-card_title fr-title/a.fr-card_link
```

```
https://www.justice.gouv.fr/prise-charge-victimes-d'infractions-penales
```

```
/html/body.path-frontpage page-node-type-homepage/[...] /article.fr-enlarge-link fr-card fr-card-detail-sm fr-card-horizontal/div.fr-card_body/div.fr-card_content/h3.fr-card_titlefr-title/a.fr-card_link
```

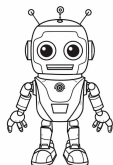
```
https://www.justice.gouv.fr/plan-d'action-justice
```

```
/html/body.path-frontpage page-node-type-homepage/[...] /div.fr-col-md-6 fr-col-12/article.fr-enlarge-link fr-card fr-card-detail-sm fr-card-horizontal/div.fr-card_body/div.fr-card_content/h3.fr-card_title fr-title/a.fr-card_link
```

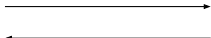
Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?
Reinforcement Learning (RL).

Leveraging **Multi-Armed Bandits** strategy [ACBF02], and especially **Sleeping Multi-Armed Bandits** [KNMS10].



Select a group
Follow a hyperlink from said group



Get a **reward**: a score describing how
“good” was this choice, to improve
future decisions ...



How many **targets** can
we reach **from this**
page?

Reward: Number of reachable, unobserved targets from the HTML page.

Exploration–Exploitation Trade-Off

At each crawling step, the choice of the group is a trade-off between:

- **Exploitation** of groups so far identified as fruitful
- **Exploration** of few times selected groups, that might hide fruitful hyperlinks.

Doing so **works well**... But requires to tell, **just given a hyperlink's URL**, if its a target or not:

- `https://www.assemblee-nationale.fr/14/pdf/ta/ta0686.pdf`
→ Easy
- `https://www.education.gouv.fr/media/90068/download`
→ Manageable
- `https://www.justice.gouv.fr/en/node/9961` → Requires learning...

Naive way: HTTP HEAD requests over the URL to get Content-Type from the header. But too costly.

Idea: Actively train a URL classifier, taking as input an URL, outputting one of the two following **classes**: **target** or **HTML**.

“on-the-fly” training allows the classifier to **adapt** to its environment and to eventual drastic changes in it.

It takes advantage of URLs that are **automatically labeled** when following a hyperlink during the crawl.

Scope:

- 18 websites expected to host **statistical datasets**
- >22M distinct pages collected

Provider types:

- **National statistical agencies:** e.g., INSEE, ABS
- **International bodies:** e.g., OECD, World Bank, UN agencies
- **Smaller, highly specialized entities:** e.g., OKFN

Diversity:

- Monolingual & multilingual websites (>30 languages)
- Websites can be:
 - Fully crawled
 - Partially crawled (limits: 1M visited pages or 3 weeks)

Dataset Heterogeneity

Website size & target density:

- Website size ranges from 4k pages (qa) to $\gg$ 1M pages (ce, il, jp)
- Targets density ranges from 2.49% (in) to 66.78% (cl)

Target exposure:

- Low exposure: 0.7% (ok) $\Rightarrow$ targets linked from very few pages
- High exposure: 41% (is) $\Rightarrow$ nearly 1 out of 2 pages links to a target

Target depth:

- Shallow targets: $\sim$ 3 clicks (cl, qa)
- Deep targets: $>$ 60 clicks (in, ju)

Highly heterogeneous dataset in both **content** (topics, languages) and **structure** (size, density, exposure, depth)

Crawlers (1/2)

- ① SB-CLASSIFIER: Our approach with URL classifier
 - ② SB-ORACLE: Our approach with an **oracle** (perfect, costless URL classifier)
 - ③ RANDOM: Random crawler
 - ④ BFS: Breadth-First Search
 - ⑤ DFS: Depth-First Search
 - ⑥ FOCUSED: A standard focused crawler (focus = targets), **without topical features**:
 - approx. depth of the hyperlink
 - 2-gram representation of the hyperlink's URL
 - 2-gram representation of the hyperlink's anchor
- **ablated version** of our approach: neither tag path structure information, nor RL

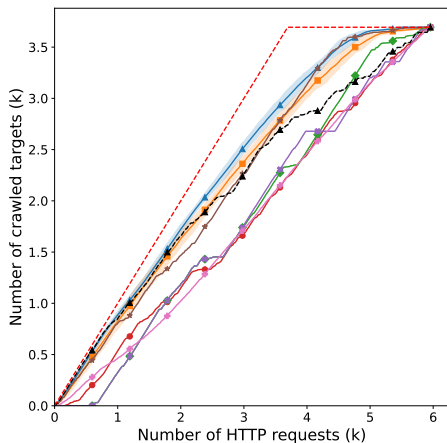
Crawlers (2/2)

- ⑦ TP-OFF: Offline-Trained, Tag Path-Based Crawler; an adaptation of ACEBOT [FS15] to target retrieval
 - Crawls 3k first pages with BFS & learns tag paths groups on this subset → requires to compute benefit of each HTML page: we provide it at no cost with an oracle (unfair advantage)
 - Then matches hyperlinks to existing tag path groups (ordered by likelihood), or given a score of 0 if too distant→ ablated version of our approach: learning offline instead of online (closest to our approach)

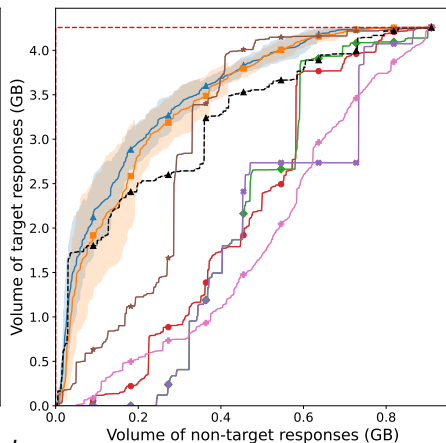
- ⑧ TRES [KKP⁺21]: Topical, RL-Based Focused Crawler; using a Bi-LSTM classifier to assess topic relevance
 - requires keywords and sample pages to define the topic: we provide it with 74 terms about the topic “target retrieval” at no cost
 - requires positive examples of fruitful HTML pages: we provide it with 1k such pages, giving partial access to the solution at no cost
 - can only handle URLs to HTML pages in its frontier: we give access to a perfect URL classifier (oracle) at no cost

Results (Overview, 1/4)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIANT



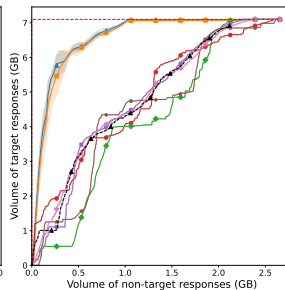
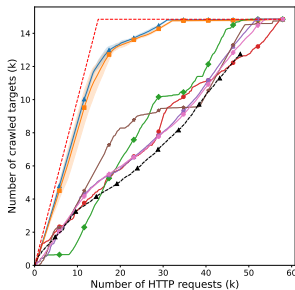
cl



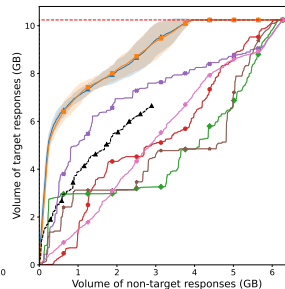
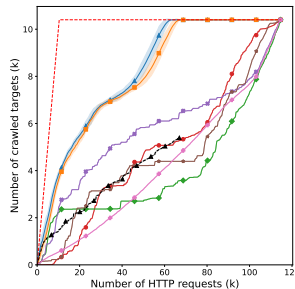
Results (Overview, 2/4)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIANT

ju:



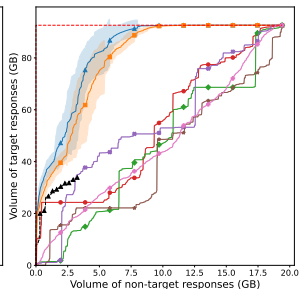
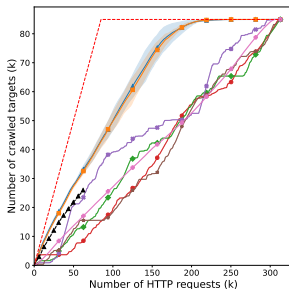
ed:



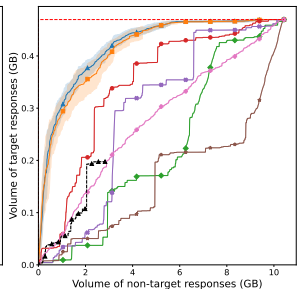
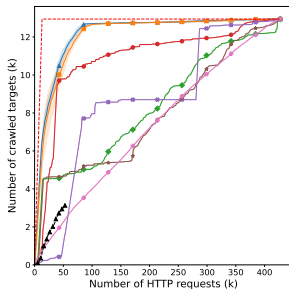
Results (Overview, 3/4)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIENT

nc:



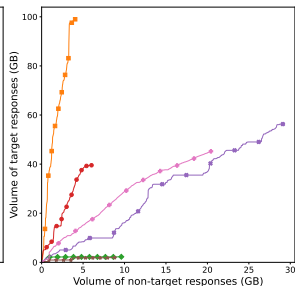
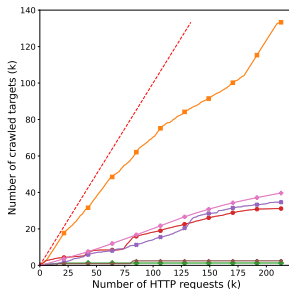
ok:



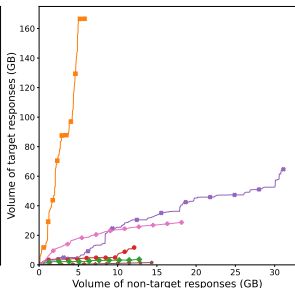
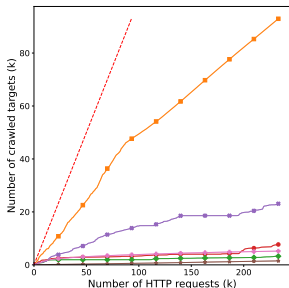
Results (Overview, 4/4)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIANT

wh:



wo:



Takeaways

- Most applications are based on Web-published data: finding them is not easy. Exhaustive methods are too costly.
- Finding an optimal crawl of a website is intractable → need of a heuristic technique.
- We presented a novel crawling technique based on reinforcement learning to dynamically learn interesting parts of the Website, that are rich in targets.
- We equip our crawler with a URL classifier that is dynamically trained with automatically-generated data, giving the class (HTML or target) of an URL at no extra cost.
- The crawler outperforms standard baselines and three adapted from the literature, sometimes reaching more than 90% of targets within less than 10% of the webpages.

Thank you for your attention!
Any questions?

Code and dataset information

https://github.com/AntoineGauquier/efficient_crawler_for_scalable_web_data_acquisition/

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Search Engines (1/3)

Why don't you just Google it?

Examples based on [Google](#) services (conducted early 2025):

- ❶ Classical [Google Search Engine](#) (and its [Programmable Search Engine](#)¹), by filtering with `site:` and `filetype:`
 - For `site:https://www.justice.gouv.fr/`
 - on `filetype:pdf` → 302 results, while > 9000 exist
 - on `filetype:tsv` → filetype is [not even recognized by GS](#)
 - For `https://www.ilo.org/`
 - on `filetype:pdf` → 641 results, while > 41k exist
 - on `filetype:zip` → 0 result, while > 2200 exist

¹<https://programmablesearchengine.google.com/>

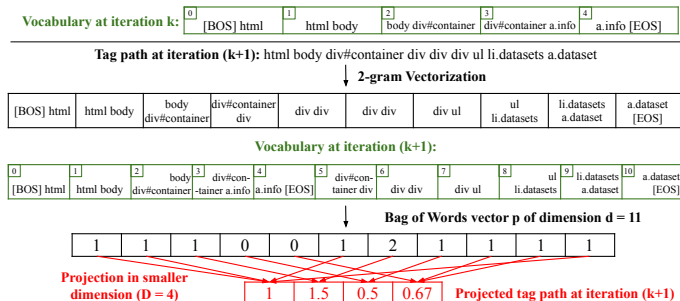
- ② Google Dataset Search²: aims at **target search**, but...
 - For site:https://www.justice.gouv.fr/ → 109 tabular data files, out of > 1100
 - For https://www.ilo.org/ → 93 datasets, out of > 170k
 - For https://www.census.gov/ → 312 datasets, out of > 800k
- ③ Google Public Data Explorer: aims at **public data exploration**, but...
 - build for human readers → only allows **manual exploration**
 - ranges over a **closed set of providers**, missing huge ones (US and UN agencies for instance)

²<https://datasetsearch.research.google.com/>

- Do not fully index any given website, and anyway limit available results to 1000.
- Don't offer any transparency or control over their process.

We devise a method that retrieves targeted data files within a budget, with hopefully **fully explainable decisions**.

Building Groups of Hyperlinks – Merging strategy



Merging strategy: all groups are represented with a **centroid**. For each tag path:

- 1 Compute its set of (approx.) k -NN, and an associated **similarity score**
- 2 If the top-score $> \theta$ (threshold): **merge** and update the **centroid**
- 3 Otherwise: create a **new group**

Dataset: Selected Websites and Characteristics (1/2)

Starting URL		Multilingual Fully Crawled	
<i>ab</i>	https://www.abs.gov.au/	X	X
<i>as</i>	https://www.assemblee-nationale.fr/	X	X
<i>be</i>	https://www.bea.gov/	X	✓
<i>ce</i>	https://www.census.gov/	X	✓
<i>cl</i>	https://www.collectivites-locales.gouv.fr/	X	✓
<i>cn</i>	https://www.cnis.fr/	X	✓
<i>ed</i>	https://www.education.gouv.fr/	X	✓
<i>il</i>	https://www.ilo.org/	✓	X
<i>in</i>	https://www.interieur.gouv.fr/	X	✓
<i>is</i>	https://www.insee.fr/	✓	✓
<i>jp</i>	https://www.soumu.go.jp/	✓	X
<i>ju</i>	https://www.justice.gouv.fr/	X	✓
<i>nc</i>	https://nces.ed.gov/	X	✓
<i>oe</i>	https://www.oecd.org/	✓	✓
<i>ok</i>	https://okfn.org/	✓	✓
<i>qa</i>	https://www.psa.gov.qa/ar/Pages/default.aspx	✓	✓
<i>wh</i>	https://www.who.int/	✓	X
<i>wo</i>	https://www.worldbank.org/	✓	X

Dataset: Selected Websites and Characteristics (2/2)

	#Available (k)	#Target (k)	HTML with Tar. (%)	Target Depth
<i>ab</i>	952.26	263.26	8.86	8.94 ($\pm$ 2.56)
<i>as</i>	994.36	93.56	4.34	4.47 ($\pm$ 4.23)
<i>be</i>	31.23	15.84	32.19	5.73 ($\pm$ 3.21)
<i>ce</i>	988.37	257.68	3.47	4.23 ($\pm$ 0.48)
<i>cl</i>	5.54	3.70	5.40	2.80 ($\pm$ 0.82)
<i>cn</i>	12.80	7.49	13.87	4.26 ($\pm$ 1.59)
<i>ed</i>	102.71	10.47	3.95	11.89 ($\pm$ 13.22)
<i>il</i>	993.92	26.56	2.53	9.63 ($\pm$ 9.83)
<i>in</i>	922.46	22.98	1.54	66.94 ($\pm$ 39.43)
<i>is</i>	285.55	168.88	41.34	5.20 ($\pm$ 1.81)
<i>jp</i>	998.26	256.34	6.30	21.37 ($\pm$ 49.34)
<i>ju</i>	56.61	14.85	4.85	86.91 ($\pm$ 86.30)
<i>nc</i>	309.97	84.94	18.87	3.63 ($\pm$ 1.66)
<i>oe</i>	222.58	45.04	15.61	6.28 ($\pm$ 5.65)
<i>ok</i>	423.12	12.95	0.74	2.64 ($\pm$ 2.89)
<i>qa</i>	4.36	2.45	4.15	3.03 ($\pm$ 0.61)
<i>wh</i>	351.86	55.59	14.19	4.43 ($\pm$ 0.62)
<i>wo</i>	223.67	23.10	2.38	4.52 ($\pm$ 0.69)

URL Classifier Design

Evaluation metrics

- **Intra-website:** % of requests needed to reach 90% of targets
- **Inter-website:** Misclassification Rate (MR)

Feature sets

- **URL_ONLY:** URL features only
- **URL_CONT:** URL + anchor text + DOM path + surrounding text

Models LR: Linear Regression SVM: Support Vector Machine NB: Naive Bayes PA: Passive-Aggressive

Variant	<i>be</i>	<i>cl</i>	<i>cn</i>	<i>ed</i>	<i>in</i>	<i>is</i>	<i>ju</i>	<i>nc</i>	<i>oe</i>	<i>ok</i>	<i>qa</i>	MR
URL_ONLY-LR	82.1	75.1	71.3	53.2	11.7	76.1	36.5	52.6	60.7	15.9	62.3	2.62
URL_ONLY-SVM	82.7	75.7	71.8	63.6	11.3	76.0	37.4	52.2	63.5	16.7	61.5	2.99
URL_ONLY-NB	82.9	75.2	72.1	53.7	11.4	76.3	35.8	52.7	59.7	18.0	63.1	2.92
URL_ONLY-PA	82.3	74.4	71.7	53.3	11.1	75.8	36.7	51.6	60.5	15.9	60.9	2.56
URL_CONT-LR	82.2	74.4	71.9	54.3	11.3	76.4	37.8	52.9	64.7	16.8	60.0	5.93
URL_CONT-SVM	82.6	75.0	71.8	52.8	11.6	76.4	38.8	53.1	61.1	18.7	60.1	6.36
URL_CONT-NB	84.1	74.7	71.9	53.6	11.4	75.7	35.5	52.3	59.9	19.1	60.4	7.15
URL_CONT-PA	82.5	75.1	71.9	53.6	11.6	76.2	38.4	52.1	62.6	16.1	60.6	4.12

URL Classifier Results (Overview)

[Confusion matrix](#) of our URL classifier, on the fully-crawled websites (average over 15 runs)

True \ Predicted			
	HTML (%)	Target (%)	Neither (%)
HTML	58.04	1.37	0.00
Target	0.75	32.19	0.00
Neither	5.34	2.41	0.00

SD Retrieval Across Sample Targets

Experimental setup

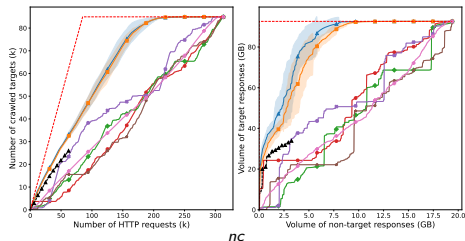
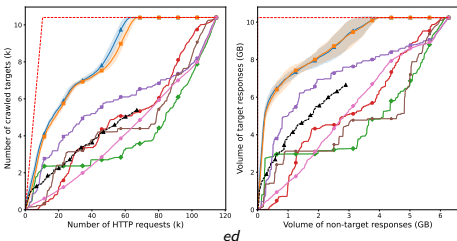
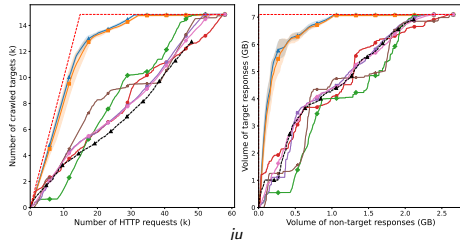
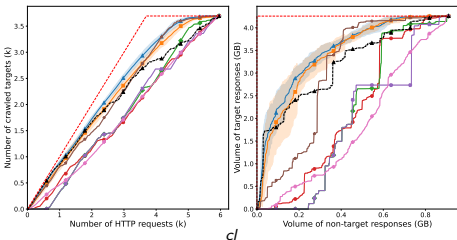
- Random sample of $7 \times 40 = 280$ targets
- 7 diverse websites

	<i>be</i>	<i>ed</i>	<i>is</i>	<i>in</i>	<i>nc</i>	<i>oe</i>	<i>wh</i>
SD Yield (%)	82	35	93	40	83	60	40
Mean # SDs / Target	9.1	2.8	2.9	2.1	2.1	4.9	1.4

- SDs are present in a substantial fraction of targets across all websites
- Even non-statistical websites show notable SD presence
- SDs are concentrated: when present, typically > 2 SDs per target

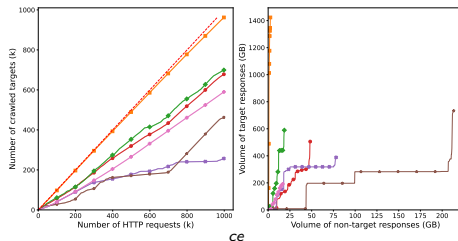
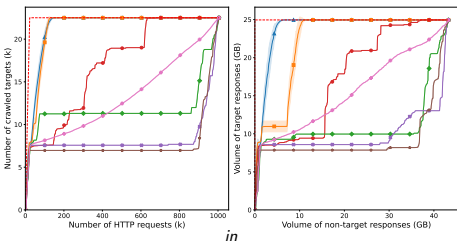
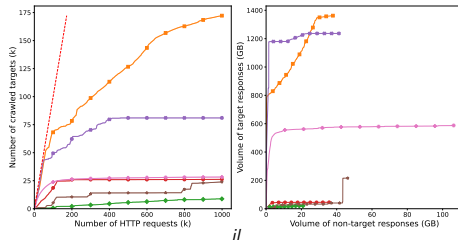
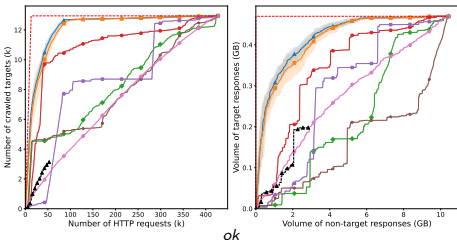
Results (Overview, 1/3)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIANT



Results (Overview, 2/3)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIENT



Results (Overview, 3/3)

SB-ORACLE SB-CLASSIFIER FOCUSED TP-OFF BFS DFS RANDOM TRES OMNISCIANT

